

Aquablend — MILP Mathematical Formulation

MILP Stream

1 Overview

Water corporations draw drinking water from multiple sources, such as reservoirs, river intakes, and groundwater bores, that differ in cost and in raw water quality (pH, alkalinity, turbidity, and so on). Before that water can be supplied for human consumption, it is treated at a plant and delivered to demand zones. The question this model answers is: which sources should be drawn from, how much from each, and how should that water be routed to plants and zones, so that demand is met, and water quality stays within safe limits, at the lowest possible cost?

This is formulated as a Mixed-Integer Linear Program (MILP). Source, plant, removal-process, and chemical-dosing activation decisions are binary, while water flows, removal loads, and chemical masses are continuous. Transfer links are represented directly by continuous flow variables and capacity limits because the current formulation does not assign fixed activation costs to arcs. The objective balances fixed activation costs against the variable cost of drawing, transferring and treating water.

Treatment is modelled in three parts. A flat rate per megalitre (ML) covers the cost of operating a plant, each removal process is charged based on the load it removes, and chemical dosing is charged based on the mass of chemical used. The load-based removal cost and chemical dosing cost make source blending an economic decision rather than purely a feasibility problem: a cheaper but more contaminated source may incur higher treatment costs than a cleaner, more expensive source. Removal processes reduce parameter loads, while chemical dosing can add load for supported quality parameters where a defensible linear conversion is available. This allows the model to represent both removal of excessive constituents and chemical adjustment used to meet required lower quality limits.

The toy case considers the simplest version of this problem: one treatment plant and one demand zone, fed by several candidate sources. With only one plant and one zone, there is no routing choice to make: every active source sends its full withdrawal to the plant, and the plant sends its entire outflow to the demand zone. As a result, the routing variables are determined by the source withdrawals, leaving the model to determine which sources to activate, how much to draw from each, and any required treatment decisions. Section 7 derives this toy case as a special case of the general network formulation, so the same formulation can later be extended to multiple plants and zones without being rebuilt from scratch.

2 Notation

Convention. This is a single-period model (one representative day), so no day index is used. Parameters are uppercase; decision variables are lowercase, using Greek letters for binary decisions and lowercase Latin for continuous quantities. A subscript is an index; an overbar/underbar is an upper/lower bound.

2.1 Sets

Table 1: Notations on sets of input parameters

Notation	Description
$\mathcal{S}$	set of water sources, $s \in \mathcal{S}$
$\mathcal{T}$	set of treatment plants, $t \in \mathcal{T}$

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Notation	Description
$\mathcal{Z}$	set of demand zones, $z \in \mathcal{Z}$
$\mathcal{P}$	set of water quality parameters, $p \in \mathcal{P}$
$\mathcal{R}$	set of removal processes, $r \in \mathcal{R}$
$\mathcal{R}_t$	all removal processes installed at plant t regardless of what parameters they act on, $\mathcal{R}_t \subseteq \mathcal{R}$
$\mathcal{R}_{tp}$	removal processes installed at plant t acting on parameter p , as an ordered sequence $r_1, \dots, r_{n_{tp}}$ drawn from $\mathcal{R}_t$, in the order water flows through them
$\mathcal{K}$	set of treatment chemicals and additives, $c \in \mathcal{K}$
$\mathcal{K}_t$	subset of treatment chemicals and additives available for dosing at plant t , with $\mathcal{K}_t \subseteq \mathcal{K}$

$n_{tp} = |\mathcal{R}_{tp}|$ is written with the index because plants differ in how many processes they run, and how many processes in a plant act on a parameter. For example, a plant with three processes removing alkalinity has $n_{t,\text{alk}} = 3$, while one with a single removal process diminishing turbidity alone has $n_{t,\text{turb}} = 1$.

The plant-specific set $\mathcal{K}_t$ represents chemical availability at plant t . Unlike the parameter-specific ordered removal sequence $\mathcal{R}_{tp}$, $\mathcal{K}_t$ is an unordered subset containing only the chemicals and additives that can be dosed at plant t .

2.2 Parameters

Table 2: Notations on input parameters

Notation	Description	Units
<i>Demand</i>		
D_z	volume of water to deliver to demand zone z	ML/day
<i>Sources</i>		
F_s	fixed cost of activating source s	\$/day
C_s	unit cost of drawing water from source s	\$/ML
$\underline{W}_s$	minimum daily withdrawal from source s	ML/day
$\overline{W}_s$	maximum daily withdrawal from source s	ML/day
<i>Treatment plants</i>		
F_t	fixed cost of activating treatment plant t	\$/day
C_t	unit treatment cost at plant t	\$/ML
$\underline{V}_t$	minimum daily throughput of plant t	ML/day
$\overline{V}_t$	maximum daily throughput of plant t	ML/day
<i>Arcs</i>		
$\overline{L}_{st}$	maximum flow on the arc from source s to plant t	ML/day
$\overline{L}_{tz}$	maximum flow on the arc from plant t to zone z	ML/day
C_{st}	unit cost of transferring water from source s to plant t	\$/ML
C_{tz}	unit cost of transferring water from plant t to zone z	\$/ML
<i>Water quality</i>		
Q_{sp}	value of quality parameter p in source s	units of p
$\underline{Q}_p$	lower regulatory limit for parameter p	units of p
$\overline{Q}_p$	upper regulatory limit for parameter p	units of p
$\underline{I}_{tp}$	minimum value of p that plant t can accept	units of p

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Notation	Description	Units
$\bar{I}_{tp}$	maximum value of p that plant t can accept	units of p
<i>Removal processes</i>		
F_{tr}	fixed daily cost of operating process r at plant t	\$/day
C_{trp}	cost per unit of load of p removed by process r at plant t	\$/ (units of p · ML)
H_{trp}	maximum fraction of the load of p removable by process r at plant t , with $0 \leq H_{trp} \leq 1$	—
<i>Chemical dosing</i>		
C_{tc}^{chem}	unit purchase cost of chemical c at plant t	\$/kg
$\underline{\rho}_{tc}$	minimum dose rate when chemical c is active at plant t	mg/L
$\bar{\rho}_{tc}$	maximum dose rate of chemical c at plant t	mg/L
η_{cp}	effect on quality parameter p per unit mass of chemical c	((units of p) · ML)/kg

Costs and limits are treated as fixed. The units of a quality parameter depend on p (e.g. mg/L, NTU). Quality parameters that are not approximately linear in their natural units are represented using transformed quantities chosen to support linear blending within the MILP. For $p = \text{pH}$, the MILP uses scaled hydrogen ion concentration in nmol/L rather than raw pH units; the source values and corresponding inlet and regulatory bounds are all represented on this same scale.

The inlet limits $\underline{I}_{tp}$ and $\bar{I}_{tp}$ bound the same physical quantity as the regulatory limits $\underline{Q}_p$ and $\bar{Q}_p$, but they have different meanings. I is what a plant can accept, while Q denotes the standards that the water must meet after treatment before reaching the demand zone.

C_{trp} is charged per unit of load removed. When p is represented as a concentration, multiplying concentration by volume gives a quantity proportional to mass, subject to consistent units. For parameters such as turbidity measured in NTU, the cost is per NTU · ML, which is not a mass; however, we assume it correlates to the mass of sediment that would produce that level of turbidity.

The plant-specific chemical parameters C_{tc}^{chem} , $\underline{\rho}_{tc}$ and $\bar{\rho}_{tc}$ are defined only for $c \in \mathcal{K}_t$. The coefficient η_{cp} is defined for chemical quality pairs $c \in \mathcal{K}$ and $p \in \mathcal{P}$; unsupported linear relationships are represented by setting $\eta_{cp} = 0$. For chemical dosing, $\underline{\rho}_{tc}$ and $\bar{\rho}_{tc}$ are concentration based operating bounds, while the optimisation variable introduced below is the total chemical mass used per day. The coefficient η_{cp} converts chemical mass into the same load units used by the water-quality formulation. When chemical c has no supported linear effect on parameter p , η_{cp} is set to zero. Direct pH adjustment is not represented by a constant linear chemical effect coefficient in this formulation.

2.3 Decision variables

Table 3: Notations on decision variables

Notation	Domain	Description	Units
<i>Sources</i>			
α_s	$\{0, 1\}$	1 if source s is activated	—
a_s	≥ 0	volume drawn from source s	ML/day
<i>Treatment plants</i>			
β_t	$\{0, 1\}$	1 if plant t is activated	—
<i>Arcs</i>			
b_{st}	≥ 0	volume of water from source s to plant t	ML/day

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Notation	Domain	Description	Units
c_{tz}	≥ 0	volume of water from plant t to demand zone z	ML/day
<i>Removal processes</i>			
σ_{tr}	$\{0, 1\}$	1 if removal process r is operated at plant t	—
e_{trp}	≥ 0	load of parameter p removed by process r at plant t	(units of p)·ML/day
<i>Water quality loads</i>			
ℓ_{tp}^i	≥ 0	quality load of parameter p at treatment stage i at plant t	(units of p)·ML/day
<i>Chemical dosing</i>			
κ_{tc}	$\{0, 1\}$	1 if chemical c is actively dosed at plant t	—
m_{tc}	≥ 0	mass of chemical c dosed at plant t	kg/day

σ_{tr} and e_{trp} are defined only for $r \in \mathcal{R}_{tp}$. The load variable ℓ_{tp}^i is defined for $i = 0, \dots, n_{tp} + 1$. Stage $i = 0$ represents the incoming quality load, stages $i = 1, \dots, n_{tp}$ represent the loads remaining after the parameter-specific removal processes, and stage $i = n_{tp} + 1$ represents the final treated load after supported chemical additions. Since we have $\mathcal{R}_{tp} = (r_1, \dots, r_{n_{tp}})$ (Table 1), the process at removal stage i is r_i : variables are indexed by the process itself, so $e_{tr_i p}$ is the same variable as e_{trp} with $r = r_i$.

The chemical variables κ_{tc} and m_{tc} are defined only for $c \in \mathcal{K}_t$. This avoids introducing chemical decision variables for chemicals that are not available at plant t . The chemical mass variable m_{tc} is used instead of dose concentration as the primary optimisation variable. This keeps chemical cost and supported quality load additions linear. The corresponding dose rate in mg/L is recovered after optimisation from the solved chemical mass and plant throughput.

3 Example network structure

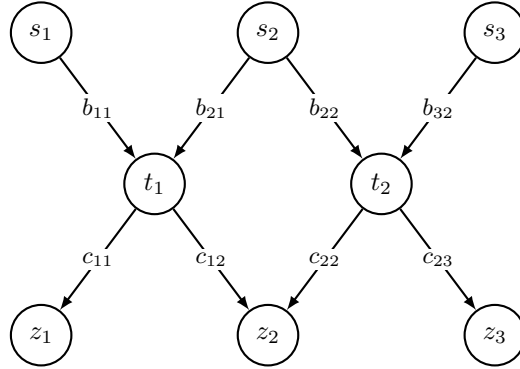


Figure 1: One example of the network structure (sources $\rightarrow$ treatment plants $\rightarrow$ demand zones). Each source-plant arc has a flow variable b_{st} ; each plant-zone arc has c_{tz} . The formulation is generic over any number of sources, plants, zones and links.

4 Objective

Minimise total cost over a single period.

Costs are grouped according to the part of the network to which they belong: sources, arcs, plants, removal processes, and chemical dosing. Sources and plants carry fixed activation costs and variable costs based on the volume handled. Removal processes carry a fixed operating cost and a variable cost based on the load removed. Chemical dosing is charged according to the mass of each chemical used, reflecting the economic trade-off between treatment performance and chemical consumption [7]. Arcs are charged on flow volume only, since the model does not currently assign a fixed cost to activating a link.

The removal and chemical costs make source blending an economic decision rather than purely a feasibility problem. A source requiring greater contaminant removal or additional chemical adjustment may incur a higher treatment cost than a source whose raw quality is naturally closer to the required outlet limits.

$$\begin{aligned}
\min \quad & \sum_{s \in \mathcal{S}} F_s \alpha_s && \text{source fixed cost} \\
& + \sum_{s \in \mathcal{S}} C_s a_s && \text{source draw cost} \\
& + \sum_{t \in \mathcal{T}} F_t \beta_t && \text{plant fixed cost} \\
& + \sum_{t \in \mathcal{T}} C_t \sum_{s \in \mathcal{S}} b_{st} && \text{plant throughput cost} \\
& + \sum_{s \in \mathcal{S}} \sum_{t \in \mathcal{T}} C_{st} b_{st} + \sum_{t \in \mathcal{T}} \sum_{z \in \mathcal{Z}} C_{tz} c_{tz} && \text{arc transfer cost} \\
& + \sum_{t \in \mathcal{T}} \sum_{r \in \mathcal{R}_t} F_{tr} \sigma_{tr} && \text{removal process fixed cost} \\
& + \sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{r \in \mathcal{R}_{tp}} C_{trp} e_{trp} && \text{removal cost} \\
& + \sum_{t \in \mathcal{T}} \sum_{c \in \mathcal{K}_t} C_{tc}^{\text{chem}} m_{tc} && \text{chemical dosing cost}
\end{aligned} \tag{1}$$

5 Constraints

Non-negativity. All continuous variables must be non-negative.

$$\begin{aligned}
a_s &\geq 0, \quad \forall s \in \mathcal{S}, \\
b_{st} &\geq 0, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \\
c_{tz} &\geq 0, \quad \forall t \in \mathcal{T}, \forall z \in \mathcal{Z}, \\
e_{trp} &\geq 0, \quad \forall t \in \mathcal{T}, \forall p \in \mathcal{P}, \forall r \in \mathcal{R}_{tp}, \\
\ell_{tp}^i &\geq 0, \quad \forall t \in \mathcal{T}, \forall p \in \mathcal{P}, i = 0, \dots, n_{tp} + 1, \\
m_{tc} &\geq 0, \quad \forall t \in \mathcal{T}, \forall c \in \mathcal{K}_t.
\end{aligned} \tag{2}$$

Demand satisfaction. The total volume of water sent from all treatment plants to demand zone z via the arcs c_{tz} must be at least the demand D_z for that zone.

$$\sum_{t \in \mathcal{T}} c_{tz} \geq D_z, \quad \forall z \in \mathcal{Z}. \tag{3}$$

Source capacity and activation. Water can only be drawn from a source if it has been activated. If source s is inactive ($\alpha_s = 0$), the draw a_s is forced to zero. If source s is active ($\alpha_s = 1$), the draw must lie between its minimum daily withdrawal $\underline{W}_s$ and maximum daily withdrawal $\overline{W}_s$.

$$\underline{W}_s \alpha_s \leq a_s \leq \overline{W}_s \alpha_s, \quad \forall s \in \mathcal{S}. \tag{4}$$

Plant capacity and activation. A plant can process water only if it has been activated. If plant t is inactive ($\beta_t = 0$), the total volume arriving from all sources is forced to zero. If plant t is active ($\beta_t = 1$), the total incoming volume must lie between its minimum daily throughput $\underline{V}_t$ and maximum daily throughput $\overline{V}_t$.

$$\underline{V}_t \beta_t \leq \sum_{s \in \mathcal{S}} b_{st} \leq \overline{V}_t \beta_t, \quad \forall t \in \mathcal{T}. \tag{5}$$

Plant flow conservation. The total volume leaving plant t toward all zones via the arcs c_{tz} equals the total volume entering it from all sources via the arcs b_{st} .

$$\sum_{z \in \mathcal{Z}} c_{tz} = \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T}. \tag{6}$$

Source flow conservation. The volume drawn from each source a_s must equal the total volume it sends to all plants via the arcs b_{st} .

$$a_s = \sum_{t \in \mathcal{T}} b_{st}, \quad \forall s \in \mathcal{S}. \quad (7)$$

Arc capacity. Each arc has a maximum flow and can carry water only if that source or plant is active. For a source-to-plant arc, the flow b_{st} is bounded by the arc capacity $\bar{L}_{st}$, the source's maximum withdrawal $\bar{W}_s$, and the plant's maximum throughput $\bar{V}_t$, and is zero unless the source is active ($\alpha_s = 1$).

$$b_{st} \leq \min \{ \bar{L}_{st}, \bar{W}_s, \bar{V}_t \} \alpha_s, \quad \forall s \in \mathcal{S}, t \in \mathcal{T}. \quad (8)$$

Similarly, the flow c_{tz} on a plant-to-zone arc is bounded by the arc capacity $\bar{L}_{tz}$ and the plant's maximum throughput $\bar{V}_t$, and is zero unless the plant is active ($\beta_t = 1$).

$$c_{tz} \leq \min \{ \bar{L}_{tz}, \bar{V}_t \} \beta_t, \quad \forall t \in \mathcal{T}, z \in \mathcal{Z}. \quad (9)$$

Inlet treatability. A plant can only accept raw water that its installed processes are built to handle. This is a property of the plant rather than a regulatory standard, which is the reason the bound includes the index t .

$$\underline{I}_{tp} \sum_{s \in \mathcal{S}} b_{st} \leq \sum_{s \in \mathcal{S}} Q_{sp} b_{st} \leq \bar{I}_{tp} \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T}, \forall p \in \mathcal{P}. \quad (10)$$

Each Q_{sp} must be a quantity that blends linearly by volume, so that $\sum_{s \in \mathcal{S}} Q_{sp} b_{st}$ represents the total load of parameter p arriving at the plant. The corresponding blended value is obtained by dividing this load by the total incoming water volume. $\underline{I}_{tp}$ and $\bar{I}_{tp}$ are held in the same units as Q_{sp} , so where Q_{sp} stores a transformed value the limits are transformed with it.

Not all plants impose limits on every parameter. When a plant t has no meaningful limit on the incoming p , the pair is neutralised rather than removed by setting

$$\underline{I}_{tp} = 0 \quad \text{and} \quad \bar{I}_{tp} = \max_{s \in \mathcal{S}} \{ Q_{sp} : \bar{L}_{st} > 0 \}, \quad (11)$$

where $\bar{I}_{tp}$ represents the maximum value of p among the sources that can reach t . Since the source values are non-negative, a volume-weighted blend cannot fall below zero or exceed the largest value among its inputs, so neither bound can be violated.

Removal process load balance. Water passes through the plant's removal processes sequentially, so each process acts on the load remaining after the preceding process. Denoting the load of p that remains after the i -th process as ℓ_{tp}^i , the load that enters the first process is what comes from the sources:

$$\ell_{tp}^0 = \sum_{s \in \mathcal{S}} Q_{sp} b_{st}, \quad \forall t \in \mathcal{T}, \forall p \in \mathcal{P}. \quad (12)$$

Each process r_i removes some of what it receives and passes on the rest:

$$\ell_{tp}^i = \ell_{tp}^{i-1} - e_{tr_{ip}}, \quad \forall t \in \mathcal{T}, \forall p \in \mathcal{P}, i = 1, \dots, n_{tp}. \quad (13)$$

The load leaving the plant is what remains after the final process, $\ell_{tp}^{n_{tp}}$.

Removal capability. Each process removes no more than a fixed fraction $H_{tr_{ip}}$ of the arriving load, based on what was passed on by the previous process rather than the total load that arrived at the plant. Modelling a removal process as a fixed fraction of the incoming load is inspired by the wastewater treatment network literature [3, 4].

$$e_{tr_{ip}} \leq H_{tr_{ip}} \ell_{tp}^{i-1}, \quad \forall t \in \mathcal{T}, \forall p \in \mathcal{P}, i = 1, \dots, n_{tp}. \quad (14)$$

Setting $H_{trp} = 0$ expresses that process r does not act on parameter p . By (2) and (13), no combination of processes can remove more than the load that arrived, and the series behaviour is enforced process by process.

Removal activation. No removal occurs unless the removal process is active. The coefficient is the maximum load the process could ever remove. The constraint forces $e_{trip} = 0$ when $\sigma_{tri} = 0$; when $\sigma_{tri} = 1$, it reduces to a valid upper bound on the maximum removable load and therefore does not further restrict the feasible removal, formal derivations for $\sigma_{tri} = 1$ is given below.

At most $\bar{I}_{tp} \bar{V}_t$ of p can reach plant t , being the highest concentration it will accept multiplied by the maximum volume it can process daily, and the process can remove at most the fraction H_{trip} of that amount.

$$e_{trip} \leq H_{trip} \bar{I}_{tp} \bar{V}_t \sigma_{tri}, \quad \forall t \in \mathcal{T}, \forall p \in \mathcal{P}, i = 1, \dots, n_{tp}. \quad (15)$$

A larger coefficient would still be valid, but it would unnecessarily expand the feasible region of the LP relaxation [2]. This is why, where a plant does not have a genuine inlet limit on p , $\bar{I}_{tp}$ is adjusted to the maximum value that the sources are capable of producing instead of being set to an arbitrarily large number.

One may notice that, $\sigma_{tri} = 1$, (15) is immediately attained:

$$e_{trip} \leq \underbrace{H_{trip} \ell_{tp}^{i-1}}_{(14)} \leq \underbrace{H_{trip} \ell_{tp}^0}_{(13)} \leq \underbrace{H_{trip} \sum_{s \in \mathcal{S}} Q_{sp} b_{st}}_{(12)} \leq \underbrace{H_{trip} \bar{I}_{tp} \sum_{s \in \mathcal{S}} b_{st}}_{(10)} \leq \underbrace{H_{trip} \bar{I}_{tp} \bar{V}_t \beta_t}_{(5)},$$

so (15) restricts the model only through $\sigma_{tri} = 0$.

Integration with the chemical treatment stage. The removal-process equations above are retained unchanged. When chemical dosing is enabled, $\ell_{tp}^{n_{tp}}$ is the load passed from the removal process sequence to the subsequent chemical treatment stage; the final plant outlet load is defined below as $\ell_{tp}^{n_{tp}+1}$.

Chemical dosing activation. Chemical availability is represented directly by the plant specific set $\mathcal{K}_t$, so no separate availability parameter is required. For each $c \in \mathcal{K}_t$, dosing can occur only when plant t is active. The binary variable κ_{tc} therefore satisfies

$$\kappa_{tc} \leq \beta_t, \quad \forall t \in \mathcal{T}, \forall c \in \mathcal{K}_t. \quad (16)$$

Chemical dose bounds. The optimisation uses chemical mass m_{tc} in kg/day as the decision variable, while plant operating limits are expressed as dose rates in mg/L. Since

$$1 \text{ mg/L} \times 1 \text{ ML/day} = 1 \text{ kg/day},$$

the dose-rate bounds can be linked directly to the plant throughput $\sum_s b_{st}$.

The upper dose rate is enforced by

$$m_{tc} \leq \bar{\rho}_{tc} \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T}, \forall c \in \mathcal{K}_t. \quad (17)$$

Chemical mass is forced to zero when dosing is inactive by

$$m_{tc} \leq \bar{\rho}_{tc} \bar{V}_t \kappa_{tc}, \quad \forall t \in \mathcal{T}, \forall c \in \mathcal{K}_t. \quad (18)$$

When a positive minimum operating dose is specified, it is enforced only when dosing is active:

$$m_{tc} \geq \underline{\rho}_{tc} \sum_{s \in \mathcal{S}} b_{st} - \bar{\rho}_{tc} \bar{V}_t (1 - \kappa_{tc}), \quad \forall t \in \mathcal{T}, \forall c \in \mathcal{K}_t. \quad (19)$$

When $\kappa_{tc} = 1$, constraints (17) and (19) reduce to the required operating dose range. When $\kappa_{tc} = 0$, constraint (18) forces $m_{tc} = 0$, while the minimum dose requirement is relaxed. If no minimum operating dose is required, $\underline{\rho}_{tc}$ is set to zero. In this case, and provided that no other chemical specific on/off condition is required, κ_{tc} may be omitted from the implementation. The binary variable κ_{tc} is retained in the general formulation because some chemicals may require a positive minimum operating dose. This allows chemical dosing to remain optional: when $\kappa_{tc} = 1$, the specified minimum dose must be satisfied, while when $\kappa_{tc} = 0$, no chemical is dosed.

For reporting after optimisation, the realised dose rate is recovered as

$$\rho_{tc}^{\text{dose}} = \frac{m_{tc}}{\sum_{s \in \mathcal{S}} b_{st}}, \quad \text{when } \sum_{s \in \mathcal{S}} b_{st} > 0. \quad (20)$$

This division is performed only in post processing and is not part of the MILP.

Chemical quality load contribution. The formulation assumes that chemical dosing occurs as a separate treatment stage following the ordered parameter specific removal sequence $\mathcal{R}_{tp}$. For each supported chemical quality relationship, $\eta_{cp}m_{tc}$ is the load of quality parameter p contributed by chemical c at plant t . The final treatment stage load extends the existing load sequence by one stage:

$$\ell_{tp}^{n_{tp}+1} = \ell_{tp}^{n_{tp}} + \sum_{c \in \mathcal{K}_t} \eta_{cp}m_{tc}, \quad \forall t \in \mathcal{T}, \forall p \in \mathcal{P}. \quad (21)$$

If $\mathcal{R}_{tp} = \emptyset$, then $n_{tp} = 0$ and the final stage becomes

$$\ell_{tp}^1 = \ell_{tp}^0 + \sum_{c \in \mathcal{K}_t} \eta_{cp}m_{tc}.$$

For chemical quality pairs without a supported linear additive relationship, $\eta_{cp} = 0$.

Outlet water quality. The water leaving each plant must remain within the regulatory limits after both removal processes and supported chemical additions have been applied. For parameter p , $\ell_{tp}^{n_{tp}+1}$ is the final treated quality load, while $\sum_s b_{st}$ is the corresponding water volume. The outlet constraint is therefore

$$\underline{Q}_p \sum_{s \in \mathcal{S}} b_{st} \leq \ell_{tp}^{n_{tp}+1} \leq \bar{Q}_p \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T}, \forall p \in \mathcal{P}. \quad (22)$$

The volume term is $\sum_s b_{st}$ rather than $\sum_z c_{tz}$; the two are equal by plant flow conservation, and using the inflow maintains consistency across all quality constraints. Removal processes can reduce excessive parameter loads, while chemical dosing can increase supported parameter loads where a valid linear conversion coefficient η_{cp} is available. This allows lower bound compliance to be achieved through a combination of source selection, removal, and chemical adjustment rather than through source blending alone.

6 Water quality formulation logic

This section explains how the generic water quality formulation is interpreted across the treatment pathway. The model distinguishes three stages: the quality of the source blend entering a plant, the load remaining after the ordered parameter specific removal sequence $\mathcal{R}_{tp}$, and the final treated load $\ell_{tp}^{n_{tp}+1}$ after supported chemical additions. This separation allows plant treatability limits to be applied before treatment and regulatory limits to be applied after treatment.

6.1 Inlet treatability and outlet water quality

Constraint (10) limits the quality of the incoming blend to the range that plant t can accept. For each $p \in \mathcal{P}$, the incoming quality load is

$$\sum_{s \in \mathcal{S}} Q_{sp}b_{st},$$

while the total incoming water volume is

$$\sum_{s \in \mathcal{S}} b_{st}.$$

The constraint is written in load form to avoid dividing by a decision variable. It is equivalent to constraining the volume weighted average inlet quality between $\underline{I}_{tp}$ and $\bar{I}_{tp}$.

The incoming load then passes through the ordered parameter specific removal sequence $\mathcal{R}_{tp}$ according to (12)–(13). After the final removal stage relevant to parameter p , the remaining load is $\ell_{tp}^{n_{tp}}$. If $\mathcal{R}_{tp} = \emptyset$, then $n_{tp} = 0$ and this removal-stage load is simply the incoming load ℓ_{tp}^0 . Supported chemical contributions are then incorporated through (21), producing the final treated load $\ell_{tp}^{n_{tp}+1}$. The final outlet quality is constrained by (22). Therefore, the model distinguishes plant inlet treatability from final regulatory compliance: the incoming blend need only be treatable by the plant, while the treated water leaving the plant must satisfy the regulatory limits.

6.2 Turbidity

Turbidity measures the cloudiness of water and is associated with suspended particles such as sediment, organic matter, algae, clay, or runoff material. In water supply modelling, turbidity is important because high turbidity can increase treatment requirements and affect disinfection performance [14, 9, 11].

For turbidity, the quality parameter is represented by setting $p = \text{turb}$. Thus, $Q_{s,\text{turb}}$ denotes the turbidity value of source s , while $\underline{L}_{t,\text{turb}}$ and $\bar{I}_{t,\text{turb}}$ define the inlet range accepted by plant t and $\underline{Q}_{\text{turb}}$ and $\bar{Q}_{\text{turb}}$ define the required outlet range.

Turbidity is handled primarily through the removal process formulation. A process that does not remove turbidity has $H_{tr,\text{turb}} = 0$. Only the processes in $\mathcal{R}_{t,\text{turb}}$ are included in the ordered removal sequence for turbidity; installed processes that do not act on turbidity are excluded from this parameter specific sequence. This formulation does not assume that adding a coagulant produces a fixed linear reduction in turbidity; such a treatment response depends on water quality and operating conditions and therefore requires a separately justified or calibrated relationship [6]. Accordingly, it is not represented by $\eta_{c,\text{turb}}$ unless supporting data are available.

6.3 Alkalinity

Alkalinity measures the ability of water to neutralise acid [10, 8, 13]. It represents the buffering capacity of water and is relevant to pH stability, corrosion control, treatment performance, remineralisation, and water stability in distribution systems [10, 13].

For alkalinity, the quality parameter is represented by setting $p = \text{alk}$. Thus, $Q_{s,\text{alk}}$ denotes the source alkalinity. The model can represent both reductions in alkalinity through removal processes and increases in alkalinity through supported chemical additions.

The final treated alkalinity load at plant t is

$$\ell_{t,\text{alk}}^{n_{t,\text{alk}}+1} = \ell_{t,\text{alk}}^{n_{t,\text{alk}}} + \sum_{c \in \mathcal{K}_t} \eta_{c,\text{alk}} m_{tc}. \quad (23)$$

For chemicals such as sodium bicarbonate, soda ash, lime, or caustic soda, a linear alkalinity equivalent contribution can be used when a defensible equivalent weight conversion is available. The final alkalinity must then satisfy the same outlet quality constraint (22) with $p = \text{alk}$.

6.4 pH transformation

Unlike alkalinity and turbidity, raw pH is not represented as a linearly blended concentration. pH is a logarithmic measure of hydrogen ion concentration and is defined as [12]

$$\text{pH} = -\log_{10}([\text{H}^+]), \quad (24)$$

where $[\text{H}^+]$ represents hydrogen ion concentration expressed in mol/L in the defining relationship [1].

Raw pH values are therefore transformed during preprocessing. To maintain numerical consistency with the implementation, the transformed hydrogen ion concentration used by the MILP is stored in nmol/L. For the pH quality parameter,

$$Q_{s,\text{pH}} = 10^{-\text{pH}_s} \times 10^9. \quad (25)$$

The factor 10^9 converts hydrogen ion concentration from mol/L to nmol/L. This scaling changes only the numerical representation of the transformed quantity and not its physical meaning.

If the acceptable raw regulatory pH range is

$$\underline{\text{pH}} \leq \text{pH} \leq \bar{\text{pH}}, \quad (26)$$

the corresponding transformed regulatory bounds, expressed in nmol/L, are

$$\underline{Q}_{\text{pH}} = 10^{-\bar{\text{pH}}} \times 10^9, \quad \bar{Q}_{\text{pH}} = 10^{-\underline{\text{pH}}} \times 10^9. \quad (27)$$

The bound direction changes because pH and hydrogen ion concentration are inversely related [5].

The plant inlet pH limits are transformed using the same scaling. If plant t accepts an incoming raw pH range

$$\underline{\text{pH}}_t^{\text{in}} \leq \text{pH} \leq \overline{\text{pH}}_t^{\text{in}}, \quad (28)$$

the corresponding inlet bounds used in constraint (10) are

$$\underline{I}_{t,\text{pH}} = 10^{-\overline{\text{pH}}_t^{\text{in}}} \times 10^9, \quad \bar{I}_{t,\text{pH}} = 10^{-\underline{\text{pH}}_t^{\text{in}}} \times 10^9. \quad (29)$$

Thus, $Q_{s,\text{pH}}$, $\underline{I}_{t,\text{pH}}$, $\bar{I}_{t,\text{pH}}$, $\underline{Q}_{\text{pH}}$, and $\bar{Q}_{\text{pH}}$ are all represented consistently in nmol/L.

In this formulation, removal processes do not act directly on pH. For every installed process,

$$H_{tr,\text{pH}} = 0 \quad \forall t \in \mathcal{T}, r \in \mathcal{R}_t,$$

so the parameter specific removal sequence is empty:

$$\mathcal{R}_{t,\text{pH}} = \emptyset, \quad n_{t,\text{pH}} = 0.$$

Consequently,

$$\ell_{t,\text{pH}}^{n_{t,\text{pH}}} = \ell_{t,\text{pH}}^0,$$

and the transformed pH load passes unchanged through the removal part of the formulation. Likewise, direct chemical adjustment of pH is not represented by the linear additive coefficient η_{cp} , so

$$\eta_{c,\text{pH}} = 0 \quad \forall c \in \mathcal{K}$$

in this formulation. Since $n_{t,\text{pH}} = 0$ and the chemical contribution is zero, the final treatment-stage load is

$$\ell_{t,\text{pH}}^{n_{t,\text{pH}}+1} = \ell_{t,\text{pH}}^1 = \ell_{t,\text{pH}}^0.$$

This avoids treating pH response to lime, caustic soda, carbon dioxide, or other chemicals as a fixed linear relationship. A future extension may introduce a separately justified piecewise linear or nonlinear pH treatment model.

6.5 Blended pH recovery after optimisation

The optimisation model constrains pH through the scaled transformed source value $Q_{s,\text{pH}}$ in nmol/L. After the solver finds the optimal source-to-plant flows, the blended hydrogen ion concentration at plant t is calculated as

$$Q_{t,\text{pH}}^{\text{blend}} = \frac{\sum_{s \in \mathcal{S}} Q_{s,\text{pH}} b_{st}}{\sum_{s \in \mathcal{S}} b_{st}}, \quad \text{when } \sum_{s \in \mathcal{S}} b_{st} > 0. \quad (30)$$

Because $Q_{t,\text{pH}}^{\text{blend}}$ is expressed in nmol/L, it is converted back to mol/L before applying the pH definition. The blended pH is therefore recovered in post-processing as

$$\text{pH}_t^{\text{blend}} = -\log_{10} (Q_{t,\text{pH}}^{\text{blend}} \times 10^{-9}) = 9 - \log_{10} (Q_{t,\text{pH}}^{\text{blend}}). \quad (31)$$

This conversion is performed only in post processing and is not part of the MILP. In this formulation, pH is not modified by either removal processes or chemical dosing, so the recovered blended pH is also the modelled plant outlet pH.

6.6 Chemical quality effect coefficients

For each supported chemical quality relationship, the coefficient η_{cp} converts the mass of chemical c used at a treatment plant into the corresponding contribution to quality parameter p . The contribution from chemical c to parameter p at treatment plant t is defined as

$$\Delta L_{tcp} = \eta_{cp} m_{tc}. \quad (32)$$

Here, m_{tc} is the mass of chemical c dosed at plant t in kg/day, while ΔL_{tcp} is the resulting contribution to the load of quality parameter p . The coefficient η_{cp} is specific to each chemical quality pair and is treated as a fixed model parameter.

The units of η_{cp} are defined so that $\eta_{cp}m_{tc}$ has the same load units as the pre-chemical removal-stage load $\ell_{tp}^{n_{tp}}$ and the final treated load $\ell_{tp}^{n_{tp}+1}$. Thus,

$$\left[\frac{(\text{units of } p) \cdot \text{ML}}{\text{kg}} \right] \left[\frac{\text{kg}}{\text{day}} \right] = \frac{(\text{units of } p) \cdot \text{ML}}{\text{day}}.$$

The value of η_{cp} is determined according to the physical relationship between the chemical and the quality parameter being represented. For direct constituent addition, the coefficient can be calculated from molecular composition and commercial product purity. For parameters such as alkalinity or hardness, an equivalent weight conversion can be used. Where the treatment response depends on additional water quality or operating conditions, a constant coefficient is used only when a defensible empirical or calibrated linear approximation is available [6].

If no supported linear additive relationship exists between chemical c and parameter p , the corresponding coefficient is set to

$$\eta_{cp} = 0.$$

This includes relationships, such as direct chemical adjustment of pH, that are not represented by the linear chemical dosing formulation.

Because η_{cp} is a fixed parameter and m_{tc} is a continuous decision variable, the relationship in Equation (32) remains linear and is therefore compatible with the MILP formulation.

6.7 Linearity assumptions for water quality

The water quality formulation relies on simplifying assumptions that preserve linearity in the MILP. The key assumptions are:

- (i) Water from selected sources is completely mixed before quality is assessed.
- (ii) The model represents a single time period, so temporal variation in water quality is not modelled.
- (iii) Source quality values are fixed input parameters for the representative period.
- (iv) Quality parameters represented directly in Q_{sp} are assumed to blend approximately linearly by volume.
- (v) Raw pH values are not averaged directly; pH is transformed to scaled hydrogen ion concentration in nmol/L before optimisation.
- (vi) Nonlinear interactions between blended source-water quality parameters are not modelled.
- (vii) Water quality is checked at each treatment plant: incoming quality is validated against $\underline{L}_{tp}$ and $\bar{I}_{tp}$, while final treated quality is validated against $\underline{Q}_p$ and $\bar{Q}_p$ using constraint (22).

These assumptions allow the general water-quality constraints to remain linear and suitable for inclusion in the MILP formulation.

6.8 Assumptions for removal processes

In practices, several designs for removal processes exist, see [?, ?]. Therefore, we shall propose the following assumptions for convenience and simplicity.

- (i) Removal processes $\mathcal{R}_{tp}$ are conducted in a sequence.
- (ii) A removal process $r_i \in \mathcal{R}_t$ can remove more than one $p \in \mathcal{P}$, appearing as $r_i \in \mathcal{R}_{tp}$ for each p .
- (iii) Every process r_i is only dependent on its preceeding process r_{i-1} , see (13).
- (iv) Maximum removal efficiency H_{trp} is known.
- (v) Uniform treatment: $\sum_{s \in \mathcal{S}} b_{st}$ receives the same treatment.
- (vi) No water loss after passing through $r_i \in \mathcal{R}_{tp}$.

- (vii) Treatment only reduces a load, lower limits met by blending alone.
- (viii) $\mathcal{R}_{tp}$ is an ordered sequence induced by $\mathcal{R}_t$.

6.9 Assumptions for chemical dosing

The chemical dosing formulation uses the following simplifying assumptions:

- Chemical mass is the primary optimisation decision and is measured in kg/day.
- Dose rate limits are specified in mg/L and are converted to equivalent daily mass bounds through plant throughput.
- Chemical dosing is optional and can occur only for chemicals in $\mathcal{K}_t$ and when plant t is active.
- Chemical additions are completely mixed with the treated plant flow before outlet quality is assessed.
- The formulation assumes that chemical dosing occurs as a separate stage following the parameter-specific removal sequence $\mathcal{R}_{tp}$.
- The coefficient η_{cp} is fixed for the representative period and already reflects any purity or equivalent conversion required by the supported relationship.
- Chemical interactions, reaction kinetics, decay, and other nonlinear dose-response behaviour are outside the MILP unless represented by a separately justified linear approximation.

The assumptions in Section 6.8 apply specifically to the removal-process subsystem. The chemical dosing stage is modelled separately after that sequence, so supported chemical additions may adjust lower-bound quality requirements without changing the removal-process equations.

7 Application to the toy model

The toy model is not a different problem but the special case of the network formulation with three sources, a single treatment plant and a single demand zone. With one plant and one zone there is no routing choice: the flow variables are fixed by conservation, while the source decisions and any applicable removal and chemical dosing decisions remain free.

Every equation of the toy model follows from the general formulation by the substitutions below. Nothing is restated here.

Substitution	Consequence
$\mathcal{T} = \{t\}, \mathcal{Z} = \{z\}$	one plant, one zone
$\beta_t = 1$	the plant is always active, so F_t is a constant and drops out
$b_{st} = a_s$	by source flow conservation
$c_{tz} = \sum_s a_s$	by plant flow conservation
$\mathcal{R}_{tp}$ and $\mathcal{K}_t$ fixed	the plant inherits its removal and chemical dosing configuration

The routing variables are therefore determined by the source withdrawals, but treatment decisions are still inherited from the general model. Where applicable, these include process activation σ_{tr} , removal loads e_{trp} , chemical activation κ_{tc} , and chemical mass m_{tc} .

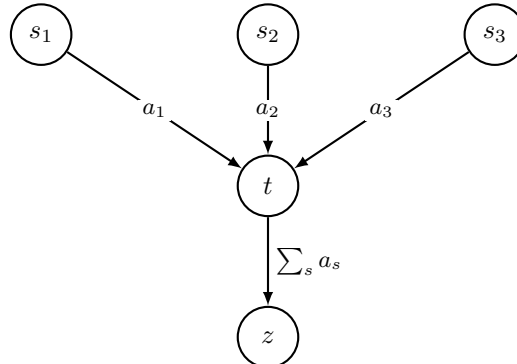


Figure 2: Toy model. Three sources draw a_1, a_2, a_3 into the single plant t , which delivers their total water volume to the single zone z .

Removal and chemical dosing change quality loads but, under the current assumptions, do not change the water volume flow balance. The outgoing water volume is therefore $\sum_s a_s$, not a quality load quantity such as the final treated load $\ell_{tp}^{n_{tp}+1}$.

Since this is just the network model with routing fixed, scaling up requires only freeing the routing variables and adding more sources, plants and zones. The same treatment and chemical dosing formulation is inherited without reformulation.

References

- [1] United States Environmental Protection Agency. pH. <https://www.epa.gov/caddis/ph>, 2026.
- [2] Jeffrey D. Camm, Amitabh S. Raturi, and Shigeru Tsubakitani. Cutting Big M Down to Size. *Interfaces*, 20(5):61–66, 1990. doi:10.1287/inte.20.5.61.
- [3] Pedro M. Castro, Henrique A. Matos, and Augusto Q. Novais. An efficient heuristic procedure for the optimal design of wastewater treatment systems. *Resources, Conservation and Recycling*, 50(2):158–185, 2007. doi:10.1016/j.resconrec.2006.06.013.
- [4] Pedro M. Castro, Joao P. Teles, and Augusto Q. Novais. Linear program-based algorithm for the optimal design of wastewater treatment systems. *Clean Technologies and Environmental Policy*, 11(1):83–93, 2009. doi:10.1007/s10098-008-0172-5.
- [5] Fondriest Environmental. pH of Water - Environmental Measurement Systems. <https://www.fondriest.com/environmental-measurements/parameters/water-quality/ph/>, 2013.
- [6] Chih-Yuan Huang, Yi-Yun Chen, Shun-Hsing Chuang, Yu-Sheng Hsu, Ray-Shyan Wu, and Kuan-Yu Ko. A precise dosing system based on a coagulation reaction model for water treatment. *Journal of Water Process Engineering*, 70:106962, 2025. doi:10.1016/j.jwpe.2025.106962.
- [7] Wei Li, Junqi Wang, Huiyu Xin, Ting Li, Jinming Duan, and Dennis Mulcahy. Determination of cost-effective optimum coagulant dosage for removal of disinfection by-product precursors in water treatment based on the theory of elasticity. *Journal of Water Process Engineering*, 47:102782, 2022. doi:10.1016/j.jwpe.2022.102782.
- [8] Test Needs Pty Ltd. Understanding Water Alkalinity: Importance, Standards, and Testing in Australia. <https://www.testneeds.com.au/understanding-water-alkalinity-importance-standards-and-testing-in-australia>, 2025.
- [9] T. Matos, M. S. Martins, R. Henriques, and L. M. Goncalves. A review of methods and instruments to monitor turbidity and suspended sediment concentration. *Journal of Water Process Engineering*, 64:105624, 2024. doi:10.1016/j.jwpe.2024.105624.
- [10] NHMRC,NRMMC. *Australian Drinking Water Guidelines Paper 6 National Water Quality Management Strategy*. National Health and Medical Research Council and National Resource Management Ministerial Council and Commonwealth of Australia, Canberra, 2011. Version 4.0, Updated June 2025. URL: <https://guidelines.nhmrc.gov.au/australian-drinking-water-guidelines>.
- [11] University of Minnesota. Metrics - Turbidity and Suspended Matter. <https://water.rs.umn.edu/turbidity>, 2017.
- [12] World Health Organization. pH in Drinking-water. <https://cdn.who.int/media/docs/default-source/wash-documents/wash-chemicals/ph.pdf>, 2007.
- [13] The Perfect Water. pH & Alkalinity in Drinking Water: Safe Levels Explained. <https://www.thep perfectwater.com/blog/post/a-guide-on-ph-alkalinity-and-their-impact-on-drinking-water-quality>, 2025.
- [14] World Health Organization. *Guidelines for Drinking-water Quality*. World Health Organization, Geneva, 4 edition, 2011. Fourth edition, incorporating first and second addenda. URL: <https://iris.who.int/server/api/core/bitstreams/69c17edd-ee26-425b-9d34-33799377e886/content>.

A Implicit constraints

This appendix complements Section 5 with inequalities that hold for the model but are not stated there explicitly. The arc linking constraints, formerly known as γ_{st} and δ_{tz} , are completely removed from our model; however, once incorporating transferring arcs activation cost is essential, we shall include them to the objective function (1) and re-introduce them, and possibly extra constraints on them are needed.

Closed nodes send no flow.

$$b_{st} \leq \overline{W}_s \alpha_s, \quad \forall s \in \mathcal{S}, t \in \mathcal{T}, \quad (33)$$

$$c_{tz} \leq \overline{V}_t \beta_t, \quad \forall t \in \mathcal{T}, z \in \mathcal{Z}. \quad (34)$$

By (2) and (7), $b_{st} \leq \sum_{t' \in \mathcal{T}} b_{st'} = a_s$, and $a_s \leq \overline{W}_s \alpha_s$ by (4). Likewise, (2) and (6) give $c_{tz} \leq \sum_{z' \in \mathcal{Z}} c_{tz'} = \sum_{s \in \mathcal{S}} b_{st}$, which is at most $\overline{V}_t \beta_t$ by (5). Hence a closed source or plant never sends water outwards.

Total supply covers total demand. Summing (3) over all zones and applying (6), (7) and (4) gives

$$\sum_{s \in \mathcal{S}} \overline{W}_s \alpha_s \geq \sum_{s \in \mathcal{S}} a_s = \sum_{t \in \mathcal{T}} \sum_{z \in \mathcal{Z}} c_{tz} \geq \sum_{z \in \mathcal{Z}} D_z. \quad (35)$$

Whatever the routing, the active sources must have enough combined capacity to cover total demand, so no set of sources whose maximum withdrawals sum to less than $\sum_z D_z$ can be the active set.

Blend bounds implied by outlet quality. Combining (22) with (12) gives, for every $t \in \mathcal{T}$ and $p \in \mathcal{P}$.

$$\underbrace{\underline{Q}_p \sum_{s \in \mathcal{S}} b_{st}}_{(22)} \leq \underbrace{\ell_{tp}^{n_{tp}}}_{(13)} \leq \underbrace{\ell_{tp}^0}_{(12)} = \sum_{s \in \mathcal{S}} \underbrace{Q_{sp} b_{st}}_{(12)} \quad (36)$$

This means a plant cannot correct a blend that is already below a lower limit, with only removal processes modelled; thereby, every lower regulatory limit must be met by source selection and blending, $\forall p \in \mathcal{P}$. Three cases follow.

- If $\underline{Q}_p = 0$ is implied by non-negativity and (36) never binds.
- If $\underline{Q}_p > 0$, as for alkalinity, (36) applies regardless of the processes installed at the plant, since none of them can help satisfy it.
- If no process acts on p (i.e. $\mathcal{R}_{tp} = \emptyset$ and $n_{tp} = 0$), then $\ell_{tp}^{n_{tp}} = \ell_{tp}^0$ and

$$\underline{Q}_p \sum_{s \in \mathcal{S}} b_{st} \leq \sum_{s \in \mathcal{S}} Q_{sp} b_{st} \leq \overline{Q}_p \sum_{s \in \mathcal{S}} b_{st},$$

so both limits must be met by blending alone.